Properties of Bott towers

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Outline

General Facts

Toric manifold and Characteristic function

Cohomology ring of toric manifolds

Bott towers and Bott manifolds

Quasitoric manifold with a Bott tower structure

Twist number of Bott manifolds

Twist number of Bott towers

Toric Manifold

- Toric Variety : a normal complex algebraic variety with $(\mathbb{C}^*)^n$ action having a dense orbit.
- Toric manifold : a compact non-singular toric variety.

We regard the compact torus T^n as the standard subgroup in $(\mathbb{C}^*)^n$, i.e., $T^n = \{z \in (\mathbb{C}^*)^n : |z| = 1\} \cong (S^1)^n$.

1. The orbit space of a toric manifold with T^n can be identified with the simple polytope
2. the action of T^n on a toric manifold is *locally standard*

By Davis and Januszkiewicz, we have the notion of a topological generalization by taking these two properties

- Quasitoric Manifold : a closed smooth manifold M of dim. $2n$ with a smooth $(S^1)^n$ action such that
  - the action is *locally standard*,
  - the orbit space $M/(S^1)^n$ is a simple convex polytope.

Characteristic Function

- P : simple polytope of dim n,
- $\mathfrak{F}(P) = \{F_1, \dots, F_m\}$: the set of facets of P,
- M : a quasitoric manifold over P,

$$\begin{tikzcd} M \arrow[d, "\pi"] \\ P \end{tikzcd}$$

Note that $\pi^{-1}(F_i)$ is the connected component of the space fixed by certain circle subgroup of T^n. Thus, we have

- $\lambda : \mathfrak{F}(P) \to H_2(BT) = \mathrm{Hom}(S^1, T^n) = \mathbb{Z}^n$: Characteristic Function with

$$\cap F_i \text{ is a vertex} \Rightarrow \{\lambda(F_i)\} \text{ is a basis of } \mathbb{Z}^n$$

(Equivariant) Cohomology ring

- $T \curvearrowright M$: A quasitoric manifold with characteristic map λ
- $P := M/T$: Simple polytope as an orbit space

We have a fibration $M \longrightarrow ET \times_T M \xrightarrow{\pi} BT$

(known) $H_T^*(M) := H^*(ET \times_T M) \overset{\cong}{=} \mathbb{Z}(P)$: face ring

Through $\pi^* : H^*(BT) \to H_T^*(M)$,

$$H_T^*(M) \text{ is an algebra over } H^*(BT) = \mathbb{Z}[t_1, \ldots, t_n]$$

Moreover the *Leray-Serre spectral sequence* collapses at the E_2 term. Thus,

$$H_T^*(M) = H^*(M) \otimes H^*(BT)$$

Assume $\lambda(F_i) = (\lambda_{1i}, \lambda_{2i}, \dots, \lambda_{ni})^t \in \mathbb{Z}^n$.

$$\Lambda := \begin{pmatrix} \lambda(F_1) & \cdots & \lambda(F_m) \end{pmatrix} = \begin{pmatrix} \lambda_{11} & \lambda_{12} & \cdots & \lambda_{1m} \\ \lambda_{21} & \lambda_{22} & \cdots & \lambda_{2m} \\ \dots & \dots & \ddots & \dots \\ \lambda_{n1} & \lambda_{n2} & \cdots & \lambda_{nm} \end{pmatrix}$$

Define linear forms

$$\theta_i := \lambda_{i1}v_1 + \cdots + \lambda_{im}v_m \in \mathbb{Z}[v_1, \dots, v_m]$$

where $1 \leq i \leq n$. In fact, $\theta_i = \pi^*(t_i)$.

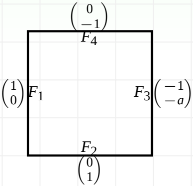
(known) $H^*(M) := H_T^*(M)/J \cong \mathbb{Z}(P)/J$ as rings,

where J is the ideal generated by $\theta_1, \dots, \theta_n$.

Hirzebruch Surface

- γ : the canonical line bundle over $\mathbb{C}P^1$
- $\underline{\mathbb{C}}$: the trivial line bundle over $\mathbb{C}P^1$

$M_a := P(\underline{\mathbb{C}} \oplus \gamma^{\otimes a})$ is a Hirzebruch surface. It indeed is a quasitoric manifold over a cube I^2 with the characteristic function



$$\Lambda = \begin{pmatrix} 1 & 0 & -1 & 0 \\ 0 & 1 & -a & -1 \end{pmatrix}$$

$$\begin{aligned} H^*(M_a) &= \mathbb{Z}[v_1, v_2, v_3, v_4]/(v_1v_3, v_2v_4, v_1 - v_3, v_2 - av_3 - v_4) \\ &\cong \mathbb{Z}[v_3, v_4]/(v_3^2, v_4^2 + av_3v_4) \end{aligned}$$

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Bott manifold

Bott tower

$$B_m \xrightarrow{\pi_m} B_{m-1} \xrightarrow{\pi_{m-1}} \cdots \xrightarrow{\pi_2} B_1 \xrightarrow{\pi_1} B_0 \;=\{\text{a point}\},$$

where $B_i = P(\eta \oplus \underline{\mathbb{C}})$ for $i = 1, \dots, m$ and η is the $\mathbb{C}$-line bundle and $\underline{\mathbb{C}}$ is the trivial line bundle over B_{i-1}.

We call each B_j a Bott manifold. Note that a Bott manifold carries a natural torus action turning it into a quasitoric manifold over a cube with

$$\Lambda = \left(\begin{array}{ccc|ccc} 1 & & 0 & -1 & & 0 \\ & \ddots & & * & \ddots & \\ 0 & & 1 & * & * & -1 \end{array}\right)$$

Cohomology ring of Bott manifold

A graded algebra S over $\mathbb{Z}$ generated by $x_1, \ldots, x_n$ of degree 2 is called a Bott quadratic algebra (BQ-algebra) over $\mathbb{Z}$ of rank n if

1. $x_k^2 = \sum_{i<k} c_{ik} x_i x_k$ where $c_{ik} \in \mathbb{Z}$ for $1 \leq k \leq n$. (In particular $x_i^2 = 0$.)
2. $\prod_{i=1}^n x_i \neq 0$.

M.Masuda, T.Panov(2007) and C, T.Panov and D.Y.Suh(2008)

- M : a quasitoric manifold over P.

$$H^*(M : \mathbb{Z}) \text{ is a BQ-algebra } \Longrightarrow P \approx I^n.$$

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$$I_P = \{v_k v_{n+k} : k = 1, \dots, n\}$$
$$J = \{v_k + v_{n+k} = \sum_{i<k} c_{ik} v_{n+i} : k = 1, \dots, n\}$$

$$H^*(M) = \mathbb{Z}[v_1, \dots, v_{2n}]/I_P + J$$

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Not all quasitoric manifolds over a cube are Bott manifolds.

Example
$\mathbb{C}P^2 \sharp \mathbb{C}P^2$ is a quasitoric manifold over I^2 with

$$\Lambda_* = \begin{pmatrix} -1 & -2 \\ -1 & -1 \end{pmatrix}.$$

Since it does not admit a complex structure, it is not a Bott manifold.

Quasitoric manifold with Bott tower structure

M.Masuda and T.Panov (2007) and N.Dobrinskaya(2001)

- M: quasitoric manifold over I^n
- $\Lambda = (E_n|\Lambda_*)$: characteristic matrix of M.

TFAE

1. M is equivalent to a Bott manifold;
2. all principal minors of $-\Lambda_*$ are 1;
3. M has a T^n-equivariant almost complex structure.

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C and D.Y. Suh

4. $H^*(M)$ is a BQ-algebra.

Generalized Bott manifolds

C, M.Masuda and D.Suh (2008)

- M: quasitoric manifold over $\prod_{j=1}^m \Delta^{n_j}$

$$H^*(M) = H^*(\prod_{j=1}^{m} \mathbb{C}P^{n_j}) \Longleftrightarrow M \cong \prod_{j=1}^{m} \mathbb{C}P^{n_j}$$

C and S. Park

- $n_1 \leq n_2 \leq \cdots \leq n_m$
- B : generalized Bott tower

$H^*(M) = H^*(B) \Longrightarrow M$ is homeomorphic to some Bott tower.

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Twist Number

A Bott tower

$$B_m \xrightarrow{\pi_m} B_{m-1} \xrightarrow{\pi_{m-1}} \cdots \xrightarrow{\pi_2} B_1 \xrightarrow{\pi_1} B_0 \; = \{\text{a point}\},$$

is called t-twisted if only t of the fibrations are non-trivial.

- M : a Bott manifold

M is called t-twisted if M is homeomorphic to B_m whose Bott tower structure is t-twisted. The minimal number of t is called twist number of M.

Lemma

A t-twisted Bott manifold M has the Bott tower structure whose only last t stages are twisted.

Cohomological complexity

Recall BQ-algebra

$$S = \mathbb{Z}[x_1, \ldots, x_m]/x_k(x_k + f_k) = 0 \text{ for } k = 1, \ldots, m,$$

where $f_k = \sum_{i<k} c_{ik} x_i$.
The number of nonzero f_k's is called the cohomological index of S. By up to graded ring isomorphism, the cohomological index can be reduced. The minimum is called the cohomological complexity of S.

Note that

the cohomological complexity of $H^*(M) \leq$ the twist number of M,

where M is a Bott manifold.

Theorem

Theorem

- M : a Bott manifold

the cohomological complexity of $H^*(M)$ $=$ the twist number of M,

Theorem

Theorem

- M : a Bott manifold

the cohomological complexity of $H^*(M) =$ the twist number of M,

Corollary 1 (agree with the result of M.Masuda and T.Panov(2007))

- M : a quasitoric manifold

$$H^*(M) \cong H^*((\mathbb{C}P^1)^m) \Longleftrightarrow M \cong (\mathbb{C}P^1)^m$$

Corollary 2

A class of quasitoric manifolds whose cohomology ring is BQ-algebra with complexity 1 is cohomologically rigid.

Proof

Let $\{B_m\}$ be a t-twisted Bott structure of M. We may assume that fibration $B_j \to b_{j-1}$ is trivial for $j = 1, \dots, m-t$. Let s be a cohomological complexity of M. Indeed, $t \geq s$. Suppose that $t > s$. We have

$$H^*(B_m) = \mathbb{Z}[x_1, \dots, x_m]/\{x_i^2 + f_i x_i = 0\}$$

$$\text{where } f_i = \begin{cases} 0 & \text{for } 1 \leq i \leq m-t \\ \sum_{j=1}^{i-1} c_{ij} x_j & \text{otherwise} \end{cases}$$

Since the cohomological complexity of the Bott tower is s, there is an isomorphism ψ such that

$$\psi : H^*(B_m) \to \mathbb{Z}[y_1, \dots, y_m]/\{y_i^2 + g_i y_i = 0\}$$

$$\text{where } g_i = \begin{cases} 0 & \text{for } 1 \leq i \leq m-s \\ \sum_{j=1}^{i-1} d_{ij} x_j & \text{otherwise} \end{cases} .$$

Claim

$\exists\, n$ ($m - t < n < m$) s.t. $f_n \equiv 0 (\text{mod } 2)$ and $f_n^2 = 0 \in H^*(B_{n-1})$.

Hence we can write as $f_n + 2w = 0$. Consider the line bundle $\underline{\mathbb{C}} \oplus \gamma^{f_n}$ over B_{n-1}. Then
$c(\gamma^w \oplus \gamma^{f_n+w}) = (1+w)(1+f_n+w) = 1 + (f_n + 2w) - \frac{f_n^2}{4} = 1.$

Lemma

A sum of two line bundles over any Bott manifold is trivial if and only if the total Chern class is 1.

Hence $\underline{\mathbb{C}} \oplus \gamma^{f_n}$ is trivial. Since $P(\underline{\mathbb{C}} \oplus \gamma^{f_n}) \cong P(\gamma^w \oplus \gamma^{f_n+w})$, $B_n \to B_{n-1}$ is trivial fibration. Thus we can reduce twist number to $t-1$. It is contradiction to the minimality of twist number. $\square$

Thank you for listening!